

CALDWELL, TX, USA

WMO: 723761

Lat: 30.516N

Lon: 96.704W

Elev: 391

StdP: 14.49

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 23901

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.7	30.3	13.9	11.3	37.2	18.3	14.0	41.8	23.8	52.1	20.9	54.8	4.7	340	0.416

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.5	99.3	73.3	97.3	73.7	95.2	74.3	79.0	85.5	78.3	85.8	77.6	86.0	8.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.4	144.8	80.6	76.6	140.8	80.2	75.3	134.7	79.6	42.6	84.9	41.9	87.5	41.3	85.3	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.2	18.3	16.6	DB	22.3	101.5	5.0	2.0	18.7	102.9	15.8	104.1	13.0	105.2	9.4	106.6	
			WB	19.9	79.5	4.5	1.2	16.7	80.4	14.0	81.1	11.5	81.8	8.3	82.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.9	48.6	52.8	60.7	67.3	74.8	81.7	83.5	84.6	79.5	69.1	59.1	51.6
	DBStd	14.58	9.15	10.25	9.04	7.40	5.85	3.56	3.04	3.09	4.93	8.01	9.67	9.48
	HDD50	367	135	82	22	2	0	0	0	0	0	1	28	97
	HDD65	1828	512	353	188	61	7	0	0	0	1	53	226	427
	CDD50	6888	93	161	354	521	769	950	1039	1073	886	594	301	147
	CDD65	2873	4	12	55	130	311	500	574	608	437	181	49	12
	CDH74	30052	12	41	257	925	2721	5600	6644	7415	4457	1641	297	42
	CDH80	14290	0	1	25	208	1001	2758	3426	4071	2192	562	45	1
Wind	WSAvg	6.8	7.0	7.8	8.3	8.3	7.4	6.6	6.0	5.4	5.0	5.9	6.6	6.9
Precipitation	PrecAvg	39.2	3.3	3.1	3.3	3.3	3.5	3.7	2.1	2.4	3.2	4.4	3.5	3.6
	PrecMax	57.8	12.2	9.2	7.9	11.1	6.6	13.9	5.7	6.3	8.2	13.9	9.6	10.3
	PrecMin	23.6	0.2	0.2	1.2	0.7	0.0	0.0	0.0	0.2	0.4	0.0	0.4	0.9
	PrecStd	9.0	2.5	2.3	1.6	2.2	2.1	3.1	1.6	1.8	2.2	3.5	2.6	2.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.2	78.8	82.3	90.1	93.1	99.9	100.0	101.8	99.5	92.5	84.0	79.0
		MCWB	63.2	63.7	67.6	70.4	73.5	72.5	73.9	72.4	71.7	72.3	70.8	67.2
	2%	DB	72.0	74.8	80.0	84.3	90.7	97.0	98.8	99.4	97.0	89.1	81.2	73.5
		MCWB	62.6	64.2	66.5	68.9	73.4	73.1	73.5	73.8	72.6	71.9	68.7	66.4
	5%	DB	68.0	72.1	77.0	81.8	88.5	93.4	96.7	97.4	93.2	86.0	77.4	71.9
		MCWB	61.0	64.0	66.1	68.7	73.1	73.7	73.9	74.3	72.4	71.7	67.2	64.9
	10%	DB	64.1	69.6	73.4	79.5	85.9	91.3	93.2	95.9	90.8	82.4	73.6	68.3
		MCWB	57.5	63.0	64.5	67.5	72.3	74.8	74.3	74.9	72.4	68.5	66.7	62.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.1	69.7	72.1	75.3	78.1	80.3	79.6	79.5	78.6	77.7	74.5	71.7
		MCDB	71.9	73.3	78.2	80.1	86.1	86.8	85.3	85.7	84.8	83.9	79.0	74.3
	2%	WB	65.8	68.1	70.4	73.4	76.7	78.9	78.7	78.6	77.6	76.2	72.3	69.1
		MCDB	69.4	72.1	75.1	80.5	84.7	85.6	85.4	86.2	85.4	82.6	76.7	72.5
	5%	WB	63.2	66.3	69.2	72.3	75.6	78.0	78.0	77.9	76.8	74.9	70.5	66.9
		MCDB	65.4	70.2	73.6	78.5	82.8	85.7	85.7	86.6	84.8	81.0	74.7	70.3
	10%	WB	59.8	64.0	67.5	71.0	74.7	77.0	77.3	77.2	75.9	73.3	68.6	64.2
		MCDB	63.4	66.7	72.3	76.6	81.5	85.1	85.7	86.5	83.9	78.8	73.6	66.5
Mean Daily Temperature Range	5% DB	MDBR	19.9	18.9	20.2	20.5	18.8	19.4	19.7	21.5	20.9	22.3	21.2	19.8
		MCDBR	23.6	21.0	20.0	21.0	20.6	22.2	23.5	23.6	24.1	22.8	22.3	22.7
		MCWBR	15.6	13.8	10.8	9.9	7.0	5.4	5.2	5.0	6.7	8.9	12.5	15.6
	5% WB	MCDBR	19.4	16.9	16.7	17.0	16.5	16.9	18.4	20.6	19.5	18.9	18.3	19.4
		MCWBR	16.7	13.9	11.6	9.5	7.5	6.1	5.5	5.4	6.2	9.4	12.9	16.2
Clear-Sky Solar Irradiance	taub	0.324	0.341	0.364	0.410	0.456	0.442	0.458	0.448	0.418	0.374	0.367	0.324	
	taud	2.503	2.469	2.405	2.291	2.207	2.298	2.283	2.304	2.340	2.445	2.402	2.513	
	Ebn at Noon	288	293	292	280	267	269	264	266	271	277	269	282	
	Edh at Noon	28	32	36	42	46	42	43	41	38	32	31	27	
All-Sky Solar Radiation	RadAvg	926	1049	1404	1704	1834	2045	2072	1954	1636	1351	1005	813	
	RadStd	92	198	131	131	142	150	156	107	148	180	111	102	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+155

Nomenclature: See separate page